

Spy Car Graybox Review: Path to a Great Game

Oct 2, 2026 · @BernoAlpha

Verdict

The graybox (published as "Shunt Graybox") has the right skeleton: relative thumb steering, auto-fire, a shunt, civilians, a ramp, a supply truck, a scripted first minute, procedural sound, combo, armor and a 3-2-1 resume. It is not fun yet, and the reason is concrete: **the traffic does not drive**. Every car moves backward in world space, so enemies and civilians rush at the player like oncoming obstacles instead of racing alongside, and the Bruiser can never actually hold position next to you to fight. Fix that one system and most of the game's fun becomes possible.

What I tested: I read all 573 lines of the build and ran a bot driver for 79 seconds in an iPhone 15 sized browser, logging every car's position 10 times a second.

Measured in the playtest	Result	Target
Time with no car near the player	47% of samples	under 15%
Longest gap with nothing happening	4.5 s	3 s max
Enemies wrecked	16 in 79 s	25 to 35
Ramp jumps	2 in 79 s	4 to 5
Supply trucks reached	1 (scripted)	2 to 3
Score lost to civilian penalties	1,600 of 2,550 earned	under 15%
How the Bruiser attacked	settled about 100 pt behind the car, then fell away for good	alongside, swerving into you

The five changes that will make it fun, in order:

1. **Real traffic physics.** All cars drive forward in world space at a speed relative to the player. Enemies match and chase; civilians cruise slower so you weave past them. (Section 4)
2. **Car-on-car collisions.** Shunted enemies must slam into civilians, other enemies, barrels and walls. Right now they only wreck on the rail, which makes ramming feel random. (Section 4)
3. **A full road.** Two to four cars on screen at all times, a ramp every 12 to 18 s, a supply truck every 25 s, and a weapon pickup that changes how you play. (Sections 5 and 6)
4. **Fair scoring.** A civilian penalty can never be bigger than a kill, and is never multiplied by the combo. (Section 2)
5. **Impact.** Native haptics, a real hit-stop, heavier screen shake on wrecks, and a wreck sound with bass. (Section 8)

Bugs and broken behavior

These are defects in the current code, ranked by how much they hurt the fun; fix all of the P0 items before any new content.

#	Priority	Bug	Where in the code	Fix
1	P0	Traffic drives backward. Civilians use <code>speedRel</code> of -60 to -120 and weak cars -40 to -80, added straight to world <code>y</code> , so they move toward the player faster than the road scrolls and read as wrong-way traffic	<code>spawnCar</code> , <code>updateWorld</code>	Store each car's world speed as a fraction of the player's speed (see Section 4) and move it by <code>c.speed * dt</code>

#	Priority	Bug	Where in the code	Fix
2	P0	The Bruiser cannot keep up. In <code>approach</code> it moves at 10 to 170 pt/s while the player does 420+, and in <code>tell</code> it chases the player with a lag that parks it about 100 pt behind. After <code>backoff</code> it is outrun and despawns	<code>updateWorld</code> , Bruiser state machine	Drive the Bruiser at player speed plus a closing term, aim for a slot beside the player, never let it fall behind unless wrecked
3	P0	Civilian penalty is multiplied by the combo before the combo resets, so hitting a civilian at x8 costs 1,600	<code>contact</code> calls <code>addScore(-200)</code> with combo active	Apply penalties unmultiplied: pass <code>quiet</code> or reset combo first
4	P0	Rear-ending re-triggers every frame. The car is pushed only 30 pt, the overlap persists, so 3 "rear hits" land in about 0.1 s: instant wreck plus up to 1.5 armor of damage to the player	<code>contact</code> , rear branch	Add a per-car hit cooldown (0.35 s) and push the hit car forward by its full overlap plus 10 pt
5	P0	Shunted enemies pass through other cars. Wrecks only happen on the rail or by the 50x90 pt proximity check at the moment of a wreck	<code>updateWorld</code>	Add car-to-car collision for any car with lateral speed above 120 pt/s (Section 4)
6	P1	Rail scraping pays forever. Holding the thumb past the edge scrapes and pays 10 points every 0.25 s with no risk	rails block in <code>simulate</code>	Pay graze only near hazards or enemies, or cap rail graze at 3 per second of scraping, then 0

#	Priority	Bug	Where in the code	Fix
7	P1	The supply truck also drives backward (<code>speedRel -90</code>), and the player passes through its body. The load window is 40 pt deep at a closing speed of about 510 pt/s, which a slow frame can skip	<code>spawnTruck</code> , truck loop	Truck drives forward at 85% of player speed; its body is solid; the load trigger is the rear ramp zone, checked with the swept distance traveled that frame
8	P1	Chain wrecks use <code>setTimeout(120)</code> , which runs in real time during pause and slow motion	<code>wreck</code>	Queue chain wrecks on the game clock
9	P1	Cars keep a lane index after the road narrows, so they get pinned against the rail and can self-wreck	<code>laneX</code> , <code>updateWorld</code>	Re-pick a valid lane when <code>laneCount</code> drops
10	P1	Hit flash and muzzle flash count down by a fixed 1/60 per frame, which is twice as fast on 120 Hz screens	<code>drawCar</code> , <code>drawPlayer</code>	Use <code>dt</code>
11	P2	Haptics use <code>navigator.vibrate</code> , which does nothing on iPhone	<code>buzz</code>	Native Core Haptics (Section 11)
12	P2	Letterbox bars still showed on both sides at iPhone 15 size in my test, even with the width-adapting <code>fit()</code>	<code>fit</code>	Verify on device; the stage should fill the screen edge to edge

#	Priority	Bug	Where in the code	Fix
13	P2	The special button reads "NO SPECIAL" for the first 22 seconds, which is dead UI	<code>updateSpecialButton</code>	Hide the button until the first pickup, then animate it in
14	P2	The Rajdhani font loads from Google Fonts and falls back when offline	<code><link></code>	Bundle the font with the app

Controls and car handling

The steering model is right (relative drag at 1.4x, 3 pt dead zone, 520 pt/s cap) and should stay; what is missing is a deliberate attack move, so ramming feels like a choice instead of an accident.

Keep as is

- Relative drag anchored to the car's target at touch-down (`carAnchor`). The car never jumps to the thumb.
- 1.4x thumb ratio, 3 pt dead zone, lean up to 10 degrees, 50% steering in the air.
- Auto-fire. Never ask the player to tap to shoot.

Add: the Slam (the game's signature verb)

- A fast horizontal flick of the steering thumb (at least 40 pt in under 120 ms) triggers a Slam: a 0.22 s sideways burst at 950 pt/s in the flick's direction.
- A Slam that connects shunts at 2.5x normal power and always wrecks a light car if it reaches a rail, wall or another car.
- 1.0 s cooldown, shown as a thin ring under the car that fills back up. A Slam that misses costs nothing but the cooldown.
- Big feedback on contact: 70 ms hit-stop, heavy haptic, a metal crunch with bass, sparks along the contact side, and the enemy car spins.
- This gives the player an attack button without a button: steer normally, flick to hit.

Fix: edges and limits

- Stop the target from going 30 pt past the rail (`targetX clamp`). Pushing into the rail should give a light scrape, sparks and a small speed loss, not free points (Bug 6).

- When the road narrows, ease the car inward over 0.4 s rather than letting the rail clamp it.

Tuning ranges to test (one variable per playtest session)

Parameter	Current	Test range
Thumb ratio	1.4	1.2 to 1.8
Lateral cap	520 pt/s	480 to 640 pt/s
Velocity smoothing	35 ms	25 to 60 ms
Slam speed	not built	800 to 1,100 pt/s
Slam cooldown	not built	0.7 to 1.4 s
Forward speed at start	420 pt/s	400 to 480 pt/s

Settings to expose

- Sensitivity 0.7x to 1.6x (the `input.sens` hook already exists).
- Left or right handed special button (already built; move the toggle from the game over card into Settings).
- Optional "tap left or right half to Slam" for players who find flicks hard.

Combat: traffic, enemies and collisions

Combat is fun when enemies drive like cars that want to hurt you and every hit can knock something into something else; today neither is true, so this section is the biggest piece of work.

1. Traffic model (replace `speedRe1`)

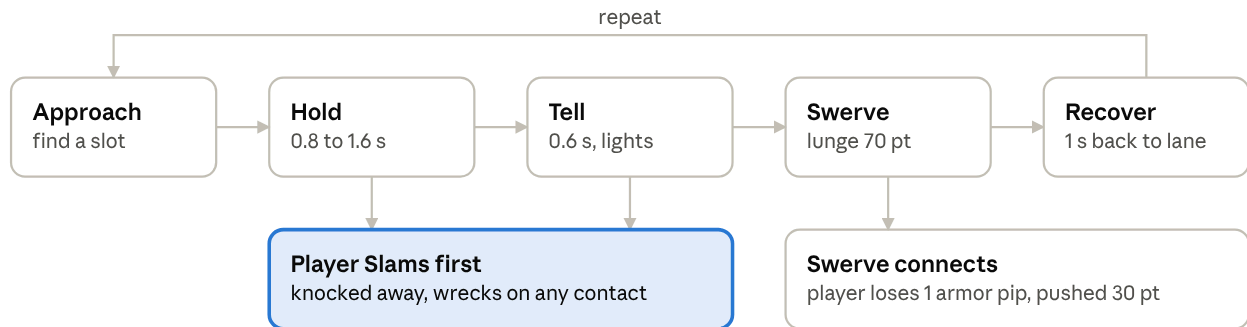
- Every car stores a world speed, `c.speed`, in pt/s and moves `c.y += c.speed * dt`, the same as the player.
- Civilians cruise at 55% to 75% of the player's speed, so they slide slowly down the screen and the player weaves past them. They change lanes occasionally (every 4 to 8 s, with a 0.6 s blinker first).
- Weak cars (gun fodder) cruise at 80% to 90% of the player's speed, so they stay in the gun's range for 1.5 to 3 s.

- Enemies (Bruisers and later types) chase: their speed is the player's speed plus a closing term toward an attack slot, capped at player speed + 180 pt/s and never less than player speed - 60 pt/s unless wrecked.
- Enemies can spawn behind the player and catch up. A rear-view cue (a red chevron at the bottom edge, 1 s before they arrive) makes this fair.

2. Bruiser behavior (rebuild)

- **Approach:** pick a slot beside the player (left or right lane, same y). Drive toward it.
- **Hold:** stay in the slot for 0.8 to 1.6 s, matching the player's lateral drift loosely. This is the window for the player to Slam first.
- **Tell:** brake lights flash and the car leans toward the player for 0.6 s (already built, keep it).
- **Swerve:** lunge sideways 70 pt at 260 pt/s. If it connects and the player is not slamming, the player loses 1 armor pip and is pushed 30 pt.
- **Recover:** drift back to its lane for 1 s, then return to Approach. Never back off and disappear.
- If the player Slams it first, it is knocked away, spins, and if it hits anything it wrecks.

The Bruiser loops until the player Slams it first



Bruiser behavior · 5 states, 2 outcomes

The Hold and Tell states are the player's window: a Slam there wins the duel, and waiting too long lets the Swerve land.

3. Car to car collisions (new)

- Give every car a mass: player 1.0, civilian 0.8, weak car 0.8, Bruiser 1.2, truck 3.0.
- Any car moving sideways faster than 120 pt/s that touches another car transfers momentum by mass. A shunted enemy that hits a civilian wrecks itself and bumps the civilian (no penalty to the player). A shunted enemy that hits another enemy wrecks

both: a Chain Wreck at 2x.

- Wrecked cars become physical debris for 1.5 s: they spin, slide and can hit other cars, which also counts as a chain.
- Cars never overlap. Resolve collisions every frame with a simple box push-out.

4. Rules for the player taking damage

- 1 armor pip lost per real hit (a Bruiser swerve that lands, a head-on with a truck, a gunner shot later). Never from the player ramming something themselves.
- 1.0 s of invulnerability after each hit, shown by the car flashing.
- Rear-ending a car costs no armor; it slows the player by 10% for 0.5 s and damages the car ahead (Bug 4 fix).

5. Civilians

- Hitting a civilian: -100 points (unmultiplied), combo breaks, the civilian bumps and honks. Never armor loss.
- Civilians are clearly slower than you and use one pastel color family, so weaving through them is a skill.
- Reward skill: passing within 12 pt of a civilian at speed scores a "Close Call" +25 and extends the combo timer by 0.5 s.

6. Graybox enemy set (do not add more until these feel great)

Enemy	Status	Job in the game
Weak car	Built	Gun fodder; proves auto-fire works
Bruiser	Built, needs the rebuild above	The ram duel; teaches the Slam
Gunner van	Add next	Sits behind, fires along a red sight line; forces lane changes
Armored truck	Add after	Immune to the base gun; must be missiled or slammed into a wall

Weapons and pickups

Today the build has one gun and one special (homing missiles from the truck, 6 shots), so the reward loop runs dry after the first truck; the graybox needs three specials and two gun upgrades to prove that collecting weapons is the hook.

What exists and what to change

- **Twin machine guns (auto-fire, 6 shots/s, 520 pt range, 26 pt lane width):** keep. Add a visible tracer line and a hit ping that rises in pitch on the shot that kills. Bullets currently pass through civilians; keep that, so auto-fire never punishes the player.
- **Homing missiles:** keep, but missiles should leave a smoke trail, arc slightly, and their explosion should splash 40 pt so one missile can chain two cars.
- **Crates (15% drop: armor or ammo):** raise to 25% and add a third type, a gun upgrade.
- **Golden crate over ramps:** keep; it is a great reason to aim jumps.

Add for the graybox (in this order)

Item	Type	Ammo	What it does	Why it is fun
Oil slick	Special	4	Drops a 50 pt slick behind; any car that crosses it spins for 1 s and slides sideways	Turns tailgating enemies into chain wrecks; works with the new chasers
Nitro	Special	3	1.5 s at +60% speed; you are invulnerable and anything you touch is wrecked	Pure power fantasy; smashing through a wall of traffic
Spread shot	Gun upgrade	until death	3-way cone, 5 shots/s	Clears wide roads; aim matters because it can't hit civilians but can push them
Heavy cannon	Gun upgrade	until death	2 shots/s, big knockback that shunts the target sideways	Shoot an enemy into a wall from range

Pickup rules

- One special at a time. A new special replaces the old one; the same special again adds ammo and levels it up (level 2 = +1 ammo and bigger effect).

- The special button is hidden until the first pickup, then flies in with a chime. Its icon shows the weapon, its rim shows the ammo as pips.
- When the player has a special and an obvious target (an armored truck, a group of 3 enemies), the button pulses once. Never more than once per target.
- Picking anything up gives a 0.15 s slow-motion beat, a chime and a light haptic. The supply truck load keeps its longer 0.8 s beat.
- Every weapon gets its own sound, muzzle flash and hit effect. If two weapons feel the same in a blind test, cut one.

Ramps, road and level flow

The jump itself is built well (1.2 s air, 0.3 s slow motion, 15% camera pull-back, road shadow, stomp, clean-landing boost), but the bot hit only 2 ramps in 79 seconds and the jumps don't jump over anything, so they feel like a bump in the road instead of a stunt.

Ramps: make every jump mean something

- Frequency: first ramp at 12 s (not 40 s), then every 12 to 18 s.
- Placement: a ramp spawns in the player's current lane or one lane over, never 3 lanes away. Today `freeLane` can put it anywhere.
- Every ramp jumps over something. Rotate these setups:
 1. **Traffic jam:** 4 to 6 civilians and an enemy bunched across the road just past the ramp. Landing on the enemy is a Stomp; jumping avoids the jam.
 2. **Broken bridge:** a 300 pt gap in the road. Missing the ramp drops you into a slower detour lane on the shoulder (lose 10% speed and the combo), never instant death.
 3. **Roadblock:** police-style barrier across 3 lanes; the ramp is the clean way over.
 4. **Golden crate in the air:** keep the current one.
- Air control: keep 50% steering. Add a landing reticle (a ring on the road where the shadow will land) during the last 0.4 s.
- Stomp: landing on an enemy is 3x points, a huge crunch, a 90 ms hit-stop and the strongest haptic in the game.
- Clean landing: keep the +120 boost and "Clean!"; add a speed-lines burst so it is felt.

Road features (add to the generator)

Feature	How often	What the player does
Explosive barrels	1 group every 20 s	Shoot them or shunt an enemy into them: blast radius 60 pt wrecks everything inside
Lane closure (cones and a merge arrow)	every 30 s	Forces traffic to bunch up, which sets up chain wrecks
Fork	every 60 to 90 s	Left branch: safer. Right branch: a ramp or the supply truck. A real choice
Concrete median wall	in long stretches	A second surface to slam enemies into
Exit and on-ramps	every 45 s	Enemies join from the side, so threats arrive from new directions

Level flow

- Shade the road in 4 "districts" that change every 60 s even in graybox (asphalt tint and rail color). Players need a sense of progress.
- Signpost the next district with a big overhead sign 3 s before it starts ("BRIDGE 0.5 MI"). This is free motivation.
- Keep the road between 3 and 5 lanes. Narrow stretches (3 lanes) should be short and intense; wide stretches (5 lanes) are where the big brawls happen.

Pacing, difficulty and the first 60 seconds

The scripted first minute is a good structure and the first kill landed at 3.5 s, but the road is empty almost half the time; the director needs a minimum on-screen population, not just a 2.8 s "nothing happened" trigger.

Fix the first 60 seconds (keep the script, tighten it)

Time	Today	Change to
0 to 3 s	"Go", empty road	Launch out of a garage ramp at 0.5 s: a jump in the first second sells the fantasy

Time	Today	Change to
3 to 8 s	One slow Bruiser alongside, ghost thumb	Keep. With the traffic fix it will really sit beside you. Show "FLICK TO SLAM" with an animated thumb
8 to 15 s	Two weak cars in your lane	Keep, plus 3 civilians to weave between so the road feels alive
15 to 22 s	Three civilians, "Spare the civilians"	Merge into the 8 to 15 s beat; use this slot for the first ramp over a small jam
22 to 32 s	Supply truck, missiles	Keep; add an armored truck at 28 s that only missiles kill
32 to 40 s	Two Bruisers	Keep; first chance at a chain wreck: place a barrel group between them
40 to 50 s	Golden crate ramp	Second ramp with a Stomp target
50 to 60 s	Mixed wave	Keep; end with an oil slick crate so the director phase starts with a new toy

Director rules (after 60 s)

- Keep a minimum population on screen: at least 2 cars at all times, at least 1 of them an enemy. Spawn ahead when below the minimum, regardless of timers.
- Run intensity in waves: 40 s of pressure (enemies up to the minute's cap), then a 10 s breather that always contains a reward (ramp, truck or crate).
- Spawn enemies both ahead and behind (behind with the rear warning chevron), never more than 2 new threats in the same 1.5 s.
- Never place two threats that need opposite moves at the same moment, and never place a ramp or crate the player can't physically reach.

Difficulty curve (target)

Minute	Enemy cap	Cars on screen	Enemy types	Speed
0 to 1	2	3 to 5	Weak, Bruiser	420 to 450 pt/s

Minute	Enemy cap	Cars on screen	Enemy types	Speed
1 to 3	3	4 to 6	+ Gunner van	450 to 520 pt/s
3 to 5	4	5 to 7	+ Armored truck, boss at minute 3	520 to 600 pt/s
5 plus	5	6 to 8	All, in combinations	up to 700 pt/s

Run length targets: a new player's first run should last 90 to 150 s; a skilled run 6 to 10 min. The bot (a poor driver) lasted 79 s, which is about right for a beginner once the grace period is tuned.

Grace period: today damage is halved for 60 s. Keep it, but also make the first death impossible before 45 s by restoring 1 armor pip at 30 s if the player is on their last pip.

Game feel: juice, audio, haptics, camera

The graybox already has more juice than most prototypes (13 procedural sounds, hit-stop, slow motion, shake, sparks, score pops, a damage vignette, smoke), so the job is to make impacts heavier and more physical rather than to add new systems.

Impacts

- **Screen shake:** today it is a random jitter of up to 20 px each frame. Replace it with directional, decaying shake: kick the camera away from the hit direction, then spring back over 150 to 250 ms. Random per-frame jitter reads as blur, not impact.
- **Hit-stop:** 60 ms on every wreck is good. Scale it: 40 ms gun kill, 70 ms Slam, 90 ms Stomp or chain wreck. Never stack beyond 120 ms.
- **Wrecks:** today a wreck is a dark box with an orange circle that spins. Make wrecks flip (scale the car's width to show it rolling), throw 3 to 5 debris chunks, leave a scorch mark on the road for 3 s, and burn for 1 s.
- **Squash and stretch:** player car squashes 10% on landing and on Slam contact.
- **Skid marks:** draw tire marks on hard lateral moves and Slams. They make speed and grip visible.
- **Speed lines:** faint streaks at the screen edges above 550 pt/s and during Nitro.

Audio (procedural is fine for graybox; plan for real assets)

- Add a bass layer (a sine drop under 80 Hz) to every wreck. Today's explosion is mostly noise, which sounds thin on phone speakers. Also add a short pitch-up "whoomp" before the bang.
- A driving music loop with 3 intensity layers tied to the director's wave, even if it is a simple procedural synth loop in graybox. Silence between sounds makes the game feel empty.
- Combo voice or stinger at x3, x5 and x8. Each tier sounds bigger.
- On iOS native: use `.ambient` audio so the game respects the silent switch and the player's own music.

Haptics (native only; the web build has none on iPhone)

Event	Core Haptics pattern
Gun hit	none (too frequent)
Gun kill	light transient
Slam contact	sharp heavy transient
Wreck / chain wreck	heavy transient plus a 120 ms rumble; chain = two pulses
Stomp	strongest transient plus 200 ms rumble
Ramp launch	rising 300 ms continuous
Landing	medium transient
Pickup	two light ticks
Damage taken	heavy, heavy
Death	long falling rumble

Camera

- Keep the car at 70% down the screen. Add a slight look-ahead toward the steering direction (up to 20 pt) so the player sees more of where they are going.
- Zoom out 5% with speed, 15% in the air (already built), and punch in 3% for 100 ms on a Stomp.

Readability, HUD and screens

The color roles are correct (cyan player, red-trimmed black enemies, pastel civilians, green truck, gold pickups, white ramps), but enemies are nearly the same value as the road, so at speed they read as holes in the asphalt rather than threats.

Readability fixes

- **Enemy contrast:** the enemy fill (#1a1b1f) on the road (#3a3d45) is too close. Keep the dark body but add a solid red roof stripe and red headlights that glow, so the red reads first. Target: an enemy is identifiable in a 0.25 s glance.
- **Weak car vs Bruiser:** today both are dark with a red outline. Make Bruisers 15% bigger with visible side rams (already drawn, make them 2x thicker) and weak cars visibly smaller and lighter gray.
- **Telegraphs:** the Bruiser tell (flashing brake lights) is too small to see at speed. Add a red arrow on the road pointing at the player during the 0.6 s tell.
- **Off-screen threats:** red chevrons at the bottom edge for enemies coming from behind; white chevrons at the top for the supply truck.
- **Damage state:** at 1 armor pip left, the player car smokes constantly and the armor bar pulses red.

HUD (top 15% only)

- Score top left (keep, 30 px is good). Add the best score in small type under it, so a new best is a visible chase.
- Armor pips: keep the three bars, but put them under the car as well during the first run, so new players connect them to the car.
- Combo: keep the center counter that grows with the multiplier, and make the timer bar thicker (8 px) and color-shift to red in its last 0.5 s.
- Special button: hidden until the first pickup (Bug 13); icon instead of the word "MISSILES".
- Callouts: keep them in the upper third and never more than 4 words while driving.

Screens

Screen	Today	Needed
Title	"Touch to drive" over the moving road	Logo, best score, "Touch to drive", a small gear for Settings, and a Daily Run button

Screen	Today	Needed
Pause	Resume, Restart, hand toggle, sound toggle	Resume (big), Restart, Settings, Quit to title. 3-2-1 on resume is already built: keep it
Game over	Cause, score, best, wrecks, peak combo, time, civilians hit; Drive Again, New Road, hand, sound	Keep the stats and the one-tap Drive Again. Add a "NEW BEST" celebration, a 1 s replay of the death, cash earned and progress toward the next unlock. Move hand and sound into Settings
Settings	none	Sound, music, haptics, sensitivity, handedness, screen shake, Reduce Motion

- Death to driving again must stay under 2 s. Today the card appears after 1.2 s and Drive Again starts immediately: good, keep it.

Progression and reasons to replay

Today the only thing that carries between runs is the best score, so after a few runs there is no reason to keep going; add a light meta loop only after the core passes the fun test, but design for it now.

Scoring (fix first, it is the backbone)

- Kills: weak car 100, Bruiser 150. Raise the Bruiser to 250 once the Slam duel exists, since it is the skill moment.
- Combo: multiplies kills, stomps and chains, up to x8, 2.5 s window. Keep. Each new kill refreshes the window.
- Penalties are never multiplied (Bug 3) and never larger than one Bruiser kill.
- Style bonuses (new): Close Call +25, Slam kill +50, enemy shunted into another enemy +100, two cars wrecked by one barrel +150.
- Distance: +1 per 10 m traveled, so survival always pays a little.

Meta loop (after the graybox gate)

1. **Cash:** 10% of score converts to cash at game over.
2. **Garage:** 3 cars at first (balanced, heavy rammer, fast and light), unlocked with cash.
3. **Upgrades:** armor, Slam power, gun damage, special capacity; 5 levels each.

4. **Starting special:** pick one unlocked special to begin each run with. This makes unlocks change how a run plays.
5. **Missions:** 3 at a time, like "Slam 5 Bruisers into the rail" or "Land 2 Stomps in one run." Each pays cash.
6. **Daily Run:** a fixed seed per day (the road generator already takes a seed), one leaderboard, a streak counter.

Moments to share

- Auto-capture the last 8 seconds of the best moment in a run (biggest chain or a Stomp) as a replay on the game over screen, with a share button. Players advertise the game for you.

iOS and performance

Keep iterating on the fun in this web graybox (it is fast to change and test on a phone), then port the proven rules to a native app; the shipping game cannot be a web page because haptics, gesture deferral and audio sessions only work natively.

While still in the web graybox

- Test on a real iPhone every session, added to the Home Screen so Safari's toolbars are gone. Desktop browser testing hides control problems.
- Move all timers to the game clock and `dt` (Bugs 8 and 10).
- Pre-render repeated gradients (explosions, ramp, glows) once instead of creating them every frame.
- Keep the 1/20 s frame cap, but run physics at a fixed 120 Hz step so collisions don't tunnel at high speed.

For the native build

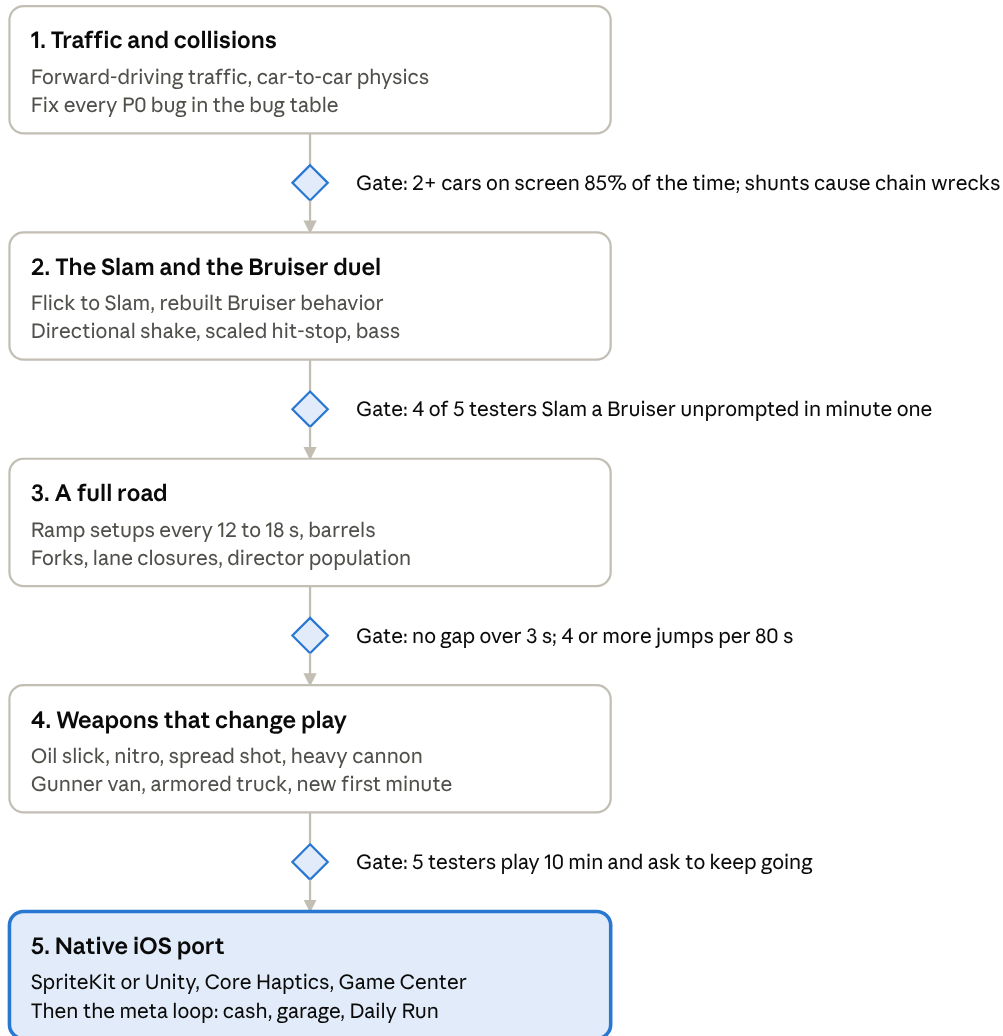
- Engine: Swift with SpriteKit is enough for this game (2.5D top-down). Unity is fine if the team knows it. Port the rules, not the canvas code.
- 60 fps locked on iPhone 12 and newer, 120 fps on ProMotion. 20 minutes of play without thermal throttling on an iPhone 13.
- Defer system edge gestures and auto-hide the home indicator so a thumb near the bottom edge never quits the game.
- Core Haptics patterns from Section 8, with a settings toggle.

- Audio session `.ambient` ; pause on interruptions (calls, Siri) and on backgrounding, resume with 3-2-1 (already built here).
- Layout fills every iPhone from SE to Pro Max and iPad with no bars: wider screens show more shoulder, not a bigger road.
- Game Center leaderboards (all-time and Daily Run), offline play, privacy-light analytics: run length, cause of death, the second a player quits their first session, weapon pick rates.
- Cold start to driving in under 5 s.

Build plan and done criteria

Build in five sprints and play-test at the end of each; a sprint is done when its gate passes on a real iPhone, not when its tasks are checked off.

Each sprint ships only when its playtest gate passes



build plan · 5 sprints, 4 playtest gates

Sprint 1 is the one that matters most: until the traffic drives forward and cars collide with each other, no amount of weapons or polish will make the game fun.

Sprint 1 checklist: traffic and collisions

- ☐ Replace `speedRe1` with a world speed per car (civilians 55% to 75%, weak cars 80% to 90%, enemies chase)
- ☐ Car-to-car collisions with mass, push-out, and chain wrecks from shunted cars
- ☐ Per-car hit cooldown on rear-ending (Bug 4)
- ☐ Unmultiplied civilian penalty of -100 (Bug 3)
- ☐ Supply truck drives forward with a solid body and a swept load trigger (Bug 7)

- ☐ Chain wrecks and flashes on the game clock (Bugs 8 and 10)
- ☐ Rail scraping capped (Bug 6)

Sprint 2 checklist: the Slam and the Bruiser

- ☐ Flick to Slam with cooldown ring
- ☐ Bruiser: approach, hold, tell, swerve, recover; never falls away
- ☐ Directional decaying shake; hit-stop scaled by event
- ☐ Bass layer on wrecks; flipping wrecks with debris and scorch marks

Sprint 3 checklist: a full road

- ☐ Director minimum population (2 cars, 1 enemy) and 40 s / 10 s waves
- ☐ Ramps every 12 to 18 s in or next to the player's lane, each over a jam, gap or roadblock
- ☐ Barrels, lane closures, forks, median wall
- ☐ Rear-threat chevrons and the Bruiser tell arrow

Sprint 4 checklist: weapons

- ☐ Oil slick, nitro, spread shot, heavy cannon
- ☐ Gunner van and armored truck
- ☐ Special button hidden until first pickup; icons instead of words
- ☐ Rewrite the first 60 seconds per Section 7

Measure every playtest (log these automatically; the bot logging in this review is a starting point)

- ☐ Share of time with 2 or more cars on screen (target 85% or more)
- ☐ Longest gap with no event (target 3 s or less)
- ☐ Wrecks per minute (target 20 to 25)
- ☐ Cause of every death (no single cause above 30%)
- ☐ Second the player quit or died in their first run (no cluster before 45 s)